Ultrafast interface magnetization dynamics in Fe/AlGaAs (001) heterostructure

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Time-resolved magnetization-induced second-harmonic generation is used to initiate and monitor coherent electronic spin precession in the Fe interface layer of a Fe/AlGaAs (001) heterostructure. The frequency, phase, and hysteretic behavior of the interface magnetization precession are found to be different from the bulk Fe film. The results indicate that faster magnetization switching can be achieved in nanostructures where interface properties dominate. © *2007 American Institute of Physics*. [DOI: 10.1063/1.2767773]

The interface magnetization plays a key role in the functioning of those spintronic devices which rely on spin-dependent tunneling for their operation. Since the source term in spin tunneling is the spin polarization of the *interface* density of states, the performance of these devices is strongly dependent on the magnetic properties of the heterointerface.[1] Magnetic metal tunnel junctions, for example, are the basis of the read heads found in hard disk drives and magnetic random access memory.[2,3] A tunnel barrier has been shown to be essential for electrical spin injection from a ferromagnetic metal into a semiconductor.[4]

A number of recent experiments have given strong evidence for noncollinearity between interface and bulk magnetization in transition metal films, with deviation angles as large as 40°–85°.[5,6] Moreover, the Fe interface magnetization in Fe/AlGaAs (001) heterostructures exhibits distinctly different switching characteristics than the bulk Fe due to interface-derived anisotropy–single step switching occurs at the interface layer, while two-jump switching occurs in the bulk Fe.[6] These results suggest that the *interface magnetization* can be manipulated and used rather than the bulk as the active layer for information storage/processing, with lower input power required for switching.

In this letter, we report on the coherent spin precession dynamics of the interface magnetization in Fe/AlGaAs (001) heterostructures using the time-resolved magnetization-induced second-harmonic generation (TR-MSHG) technique, and compare these results with the bulk spin precession dynamics obtained by time-resolved magneto-optical Kerr effect (TR-MOKE).[7] We measure the frequency, phase, and hysteretic behavior of the precession dynamics of the interface and bulk. The higher precessional frequencies observed at the interface indicate that higher speed performance can be realized in nanoscale magnetic devices where interface properties dominate.

MSHG is a nonlinear optical version of the MOKE technique that provides intrinsic interface sensitivity[8] and a very large magneto-optical response.[9] While MOKE averages the magnetic behavior over the entire film thickness, MSHG very selectively detects *only the interface contribution* due to basic symmetry constraints.[10] Several groups recently utilized the TR-MOKE technique to directly observe uniform magnetization precession in thin ferromagnetic films.[7,11,12]

Thin epitaxial Fe films with 10 nm thickness were grown at a substrate temperature of 10–15 °C by molecular beam epitaxy on $Al_{0.08}Ga_{0.92}As$ epilayers grown on GaAs (001) substrates.[13] Previous work has shown that the Fe/GaAs (001) interface is essentially abrupt and characterized by Fe–As bonding for samples grown in a similar manner in the same laboratory.[14–16] The Fe film surface was oxidized by exposure to air to suppress the MSHG signal from the surface[17] and enable measurement of the magnetization at the Fe/AlGaAs interface. This results in the growth of a 2–3 nm thick oxide layer.[13] The equilibrium magnetic properties in the bulk are verified by standard MOKE measurements and vibrating sample magnetometry. The magnetization exhibits a uniaxial in-plane anisotropy superimposed on a fourfold cubic anisotropy, leading to in-plane easy axes along ⟨100⟩ and a hard axis along the [1−10] direction.[14]

TR-MSHG and TR-MOKE experiments are performed at room temperature with a 150 fs pulsed Ti:sapphire amplifier system at 800 nm wavelength. We trigger the interface magnetization precession utilizing the excitation mechanism introduced initially for bulk spin precession in a thin ferromagnetic metal film,[11] as follows. A modulated pump beam with 50 μJ pulse energy is focused to a spot of 1.5 mm in diameter on the sample. In equilibrium, the interface magnetization is along an effective field H_{eff}. The incident pump pulse instantaneously heats up the film and induces a transient field H_{tr}. When H_{tr} has vanished ($\Delta t > 50$ ps), the vector M is rotated away from H_{eff} and starts to precess around H_{eff}. A p-polarized time-delayed probe beam with 20 μJ pulse energy is incident on the sample at an angle of 45°, as illustrated in Fig. 1(a). The reflected SHG light passes through a crossed-polarization analyzer and is detected by a photomultiplier. Similarly, the effect on the bulk magnetization is measured by a time-delayed probe beam using the TR-MOKE technique.[7]

Figure 1(b) shows the TR-MSHG signal after the excitation of Fe/AlGaAs in a magnetic field of 560 Oe applied along the in-plane hard axis [1−10] For p-in/s-out polarization combination, the oscillating signal originates from the precession of the polar (M_z) magnetization component. The Fourier transformation (see inset) yields a single frequency

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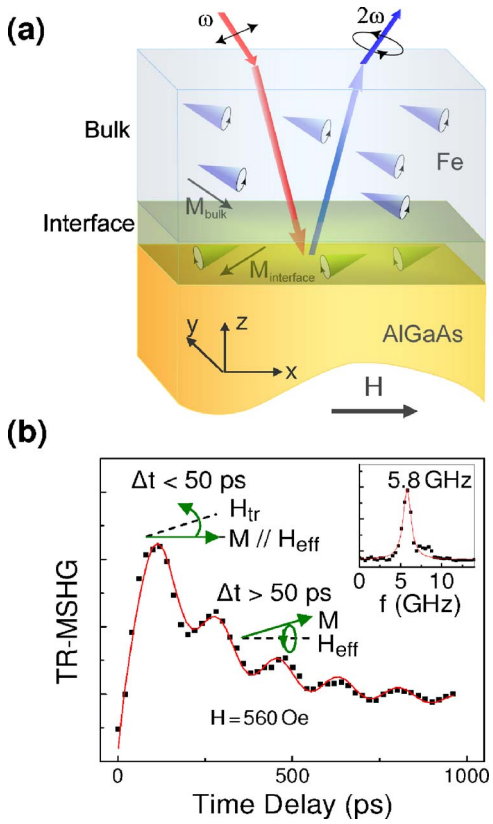

FIG. 1. (Color online) (a) Experimental configuration. (b) Time-resolved MSHG signal after excitation with a magnetic field $H=560$ Oe applied along $[1-10]$ direction. The inset shows the Fourier transformation.

of $\nu=5.8$ GHz. Figure 2(a) shows a direct comparison between interface and bulk precession for an applied field of 370 Oe. The large difference in precession frequency, 4.6 vs 7.8 GHz, suggests that the magnetic anisotropy fields are very different for the interface layer and the bulk Fe. The higher damping of bulk precession compared to interface may be caused by magnon scattering from the oxide surface. The precession frequencies at different magnetic fields are plotted in Fig. 2(b), showing a strong field dependence, which clearly distinguishes this signal from potential nonmagnetic interface contributions, e.g., phonons.

To fully understand the spin precession dynamics at the interface in Fe/AlGaAs, it is critical to obtain quantitative information on the magnetic anisotropies. The field dependence of uniform precession frequencies allows us to determine these values. From the Landau-Lifshitz-Gilbert equation, the frequencies can be described by

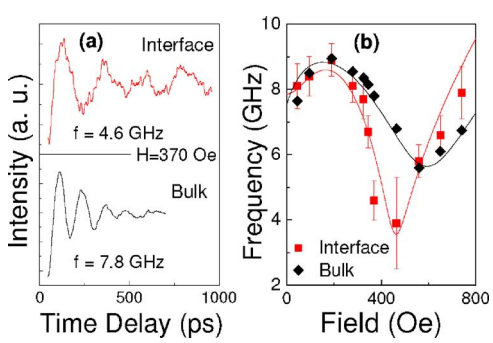

FIG. 2. (Color online) (a) Interface and bulk precession at $H=370$ Oe. (b) Precession frequency vs applied magnetic field. The solid lines are the fits by Eq. (1).

$$\omega=\gamma[(H_e\cos(\delta-\phi)+H^{\alpha})(H_e\cos(\delta-\phi)+H^{\beta})]^{1/2}, \quad (1)$$

where $H^{\alpha}=2K_{\parallel}/M_s\cos(4\phi)+2K_u/M_s\sin(2\phi)$, $H^{\beta}=H_d+H_{\perp}-K_{\parallel}/M_s\sin^2(2\phi)-K_u/M_s(\sin\phi-\cos\phi)^2$, and $\gamma=\gamma_e g/2$ (with $\gamma_e=1.76\times10^7$ Hz/Oe and $g=2.09$) is the gyromagnetic ratio. ϕ and δ are the angles between the in-plane easy axis [100] and the directions of magnetization and applied field H_e, respectively. K_u and $K_{\parallel}$ are the in-plane uniaxial and cubic anisotropy constants, respectively. H_d and $H_{\perp}$ are the demagnetization and out-of-plane anisotropy fields, respectively.

In Fig. 2, the red and black solid curves represent the best fits for the interface and bulk precession, respectively, using Eq. (1). The results reveal a distinctly different ratio of in-plane uniaxial to cubic anisotropy $r=K_u/K_{\parallel}$ between interface and bulk. A ratio of 3.6 is obtained at the interface ($K_u/M_s=172\pm28$ Oe, $K_{\parallel}/M_s=48\pm25$ Oe), while a much smaller value of 0.35 is determined for the bulk ($K_u/M_s=70\pm25$ Oe, $K_{\parallel}/M_s=200\pm20$ Oe). As a consequence, a two-step switching process in the bulk is expected from the coherent rotation model,[18] while single step switching occurs at the interface.[6] Furthermore, we find a much larger out-of-plane field $H_d+H_{\perp}$ at the interface (29 ± 5 kOe) than in the bulk (18 ± 3 kOe). The demagnetization field is assumed to be the same for the bulk and the interface, $H_d=15.7$ KOe, according to the recent report on the bulk-like magnetic moment of Fe interface layer on GaAs substrate measured by x-ray magnetic circular dichroism.[19] Hence $H_{\perp}$ is much larger at the interface than in the bulk Fe.

The large out-of-plane anisotropy at the interface may derive from a tetragonal distortion near the interface. Indeed, Gordon *et al.* found a body-centered tetragonal distortion in a 9-ML-thick Fe film on GaAs (001), which involves an in-plane contraction and an out-of-plane expansion.[20] This structural anisotropy may lead to an enhanced in-plane exchange coupling between spins at the interface layer, making the coherent rotation and uniform precession favorable over spin disorder at the interface.

Note that the sum of $H_{\perp}$ and H_d will determine the slope of the frequency-field dependence in the range of high fields where the interface precession frequency exceeds that of the bulk (Fig. 2). A departure of surface spin precession from bulk magnetization was previously observed in $Ni_{81}Fe_{19}$ films due to a reduction of H_d at the surface.[21] Here, the higher precession frequencies at the interface imply a faster switching of the interface magnetization in this thin film sample.

In Fig. 3, we further demonstrate that the interface and bulk spin precession can exhibit large phase shifts resulting from the different magnetization switchings. The hysteresis loop in Fig. 3(a) measured by static MSHG for a field applied along the $[1-10]$ direction shows a single step magnetization switching at ±4 Oe corresponding to a jump over the uniaxial hard axis $[1-10]$. Therefore, the interface magnetization is reversible in the range of positive field ($H>4$ Oe) or negative field ($H<-4$ Oe), where it stays between the easy axis [010] and the hard axis $[1-10]$ [Fig. 3(a), inset]. The red curve in Fig. 3(b) represents the interface magnetization precession dynamics measured at a field of 72 Oe sweeping down from positive high field, and the black curve is taken at the same field but sweeping up from negative high field. The two measurements show the same phase and amplitude of the magnetization precession, as expected from the

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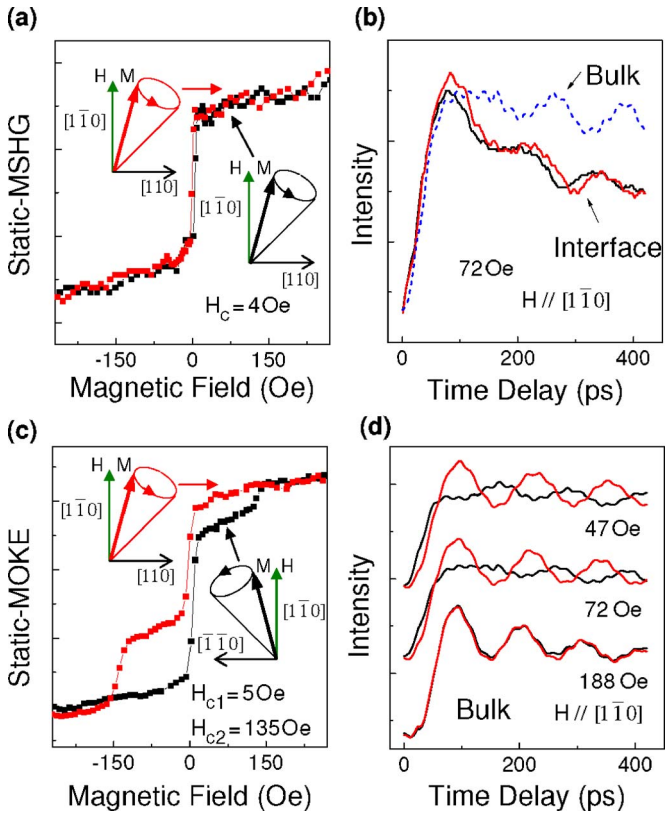

FIG. 3. (Color online) (a) Interface *M-H* loop; (b) interface precession at 72 Oe; (c) bulk *M-H* loop; and (d) bulk precessions at different fields. The black and red curves are taken with field sweeping up or down, respectively. The dashed blue curve shows bulk precession at 72 Oe with field sweeping up.

hysteresis loop that shows the same magnetization for increasing and decreasing field sweeps. The dashed blue curve shows the corresponding precession dynamics of the bulk obtained with TR-MOKE at the same field sweeping up from negative high field. The interface and bulk magnetization precessions are ~180° out of phase. This is attributed to the different switching characteristics as discussed below.

The hysteresis loop in Fig. 3(c) recorded by static MOKE shows a distinctly different reversal behavior in the bulk—a two-step switching process.[6,18,22] The first step at ±5 Oe occurs when the magnetization is pulled over the intermediate hard axis [110]. As the magnitude of the reversed applied field increases further to about ±135 Oe, the second step occurs as M rotates through the hard-hard axis [1−10].[6,18,22]

The effect of the two-step switching on the precession dynamics of the bulk is shown in Fig. 3(d). The red curves are taken at fields sweeping down, while the black curves are taken at fields sweeping up. The two curves measured at 47 Oe exhibit a phase difference close to 180°, indicating a different equilibrium magnetization direction. From the above switching model, the magnetization stays between the easy axis [010] and the hard axis [1−10] for the field sweeping down, while it stays between the easy axis [100] and the hard axis [1−10] for the field sweeping up [Fig. 3(c), inset]. Therefore, in the two measurements the magnetization stays nearly symmetric about the hard axis [1−10], i.e., the direction of the applied field. This is equivalent to a phase difference of 180° in the precession because in both cases the magnetization vector precesses counterclockwise about the equilibrium axis H_{eff} according to $\dot{M}=-\gamma M\times H_{\mathrm{eff}}$ [Fig. 3(c), inset].

Similar precession dynamics is observed when the field is tuned to 72 Oe which is between the first (H_{c1}) and second (H_{c2}) switching field [Fig. 3(d)]. However, no phase difference is observed when the field is increased to 188 Oe which is above H_{c2}. Therefore, we conclude that the two-step switching causes the observed phase shift between interface and bulk spin precession. The noncollinear spin orientation revealed in static measurements[6] is also manifested in the dynamic process on the picosecond time scale.

Our findings provide important insights into dynamic phenomena occurring at magnetic interfaces and demonstrate that the interface magnetization may be manipulated independently and at higher frequencies than that of the bulk film.

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